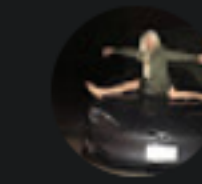


Module 6 - Data Modeling

ANALYSIS

TASK CHECKLIST

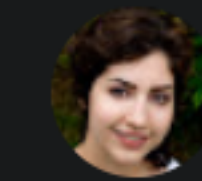
- ☒ Use requirements elicitation techniques (interview, JAD session, questionnaire, document analysis, and observation).
- ☒ Apply requirements analysis strategies as needed to discover underlying requirements.
- ☒ Develop the requirements definition.
- ☒ Develop use cases.
- ☒ Develop data flow diagrams.
- ☐ Develop entity relationship model.
- ☐ Normalize entity relationship model.



Yifei Zeng

Oct 22, 2023

On the textbook page 169, the title of this task checklist is "planning", are these tasks in the planning phase or the analysis phase?



Maryam Khezzadeh

Oct 22, 2023

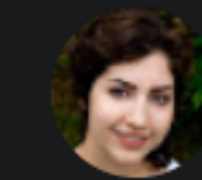
Your textbook has a mistake. These tasks are in analysis. You could figure that out by reading the first few sentences of the chapter.



David Suh

Feb 15, 2024

Marked as resolved



Maryam Khezzadeh

Feb 20, 2024

Re-opened

Blueprint for a database?

- Like building **cars** or building **houses**, building a database needs a blueprint
- We need to create a **model first** and then implement it



How do we create a model for a database?

What is a data model?

- Represents the **data** that are **used and created** by a system.
- **people, places, things** or **events** about which information is captured and how they are **related** to each other.

Logical vs. Physical Data Models

- Logical data models are **free of any implementation or technical details**
 - We are just trying to understand the data requirements of the system
- Physical data models are **more detailed and less conceptual**, addressing ways to store the data efficiently and to make the data easy to retrieve.

There are several ways to model data. One of the most commonly used techniques is **entity relationship diagramming**.

The Entity Relationship Diagram

- What exactly is a “preliminary custom order”?
- What pieces of data are included in a “drone base model” or a “drone component”?

Use Case Name: Create <u>preliminary custom drone order</u>	ID: UC-6	Priority: High
Actor: Customer		
Description: The customer selects and customizes a commercial drone to purchase		
Trigger: Customer wants to purchase a commercial drone		
Type: ☒ External ☐ Temporal		
Preconditions: 1. The customer is authenticated by logging in to his account 2. The Sales System Order Processing application is online		
Normal Course: 1.0 Order a customized drone 1. The customer selects a <u>base model drone</u> from a list of models 2. The system provides <u>availability status</u> for that model (in stock, out of stock) 3. For out of stock status, system displays expected date available a. Customer accepts future availability date; proceed to step 4 b. Customer rejects future availability date; return to step 1 4. The system displays a list of options and upgrades for the selected model 5. The customer selects desired model options and upgrades 6. Preliminary order with cost estimate is created and displayed 7. Customer may return to step 4, confirm order, save for future consideration, or exit without saving 8. Unconfirmed orders are stored in Unconfirmed Custom Order datastore 9. Confirmed orders are saved in Confirmed Custom Order datastore 10. Shop manager is notified of Confirmed Order requiring approval		
Postconditions: 1. Unconfirmed order is stored in Unconfirmed Custom Order datastore 2. Confirmed order is stored in Confirmed Custom Order datastore 3. Shop manager sent notice of Confirmed Order requiring approval		

FIGURE 4-1 Create preliminary custom drone order use case—casual format.

- What exactly is a “preliminary custom order”?
- What pieces of data are included in a “drone base model” or a “drone component”?

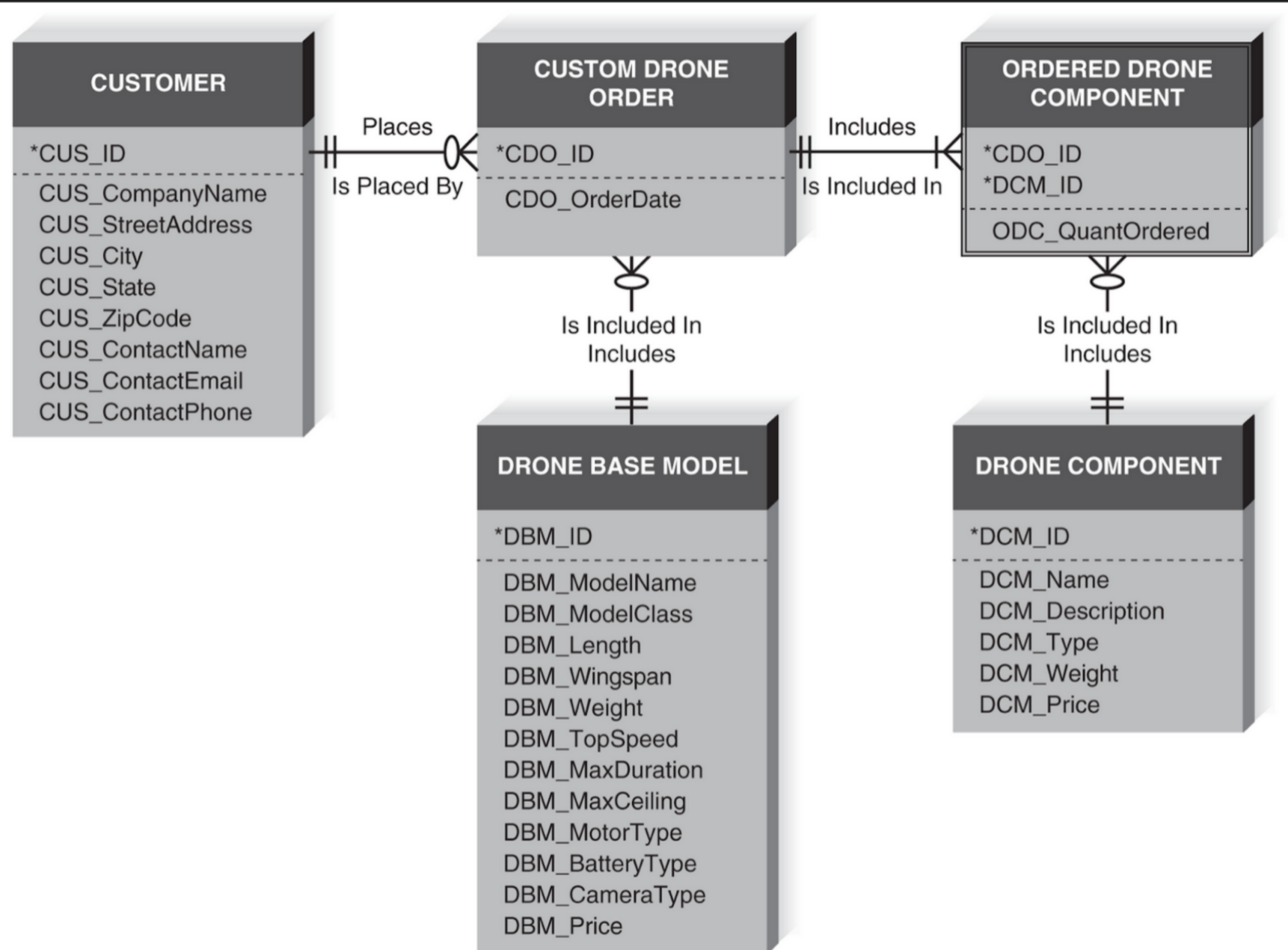


FIGURE 5-1 Partial DrönTeq sales system ERD.

- Five main categories (of data)

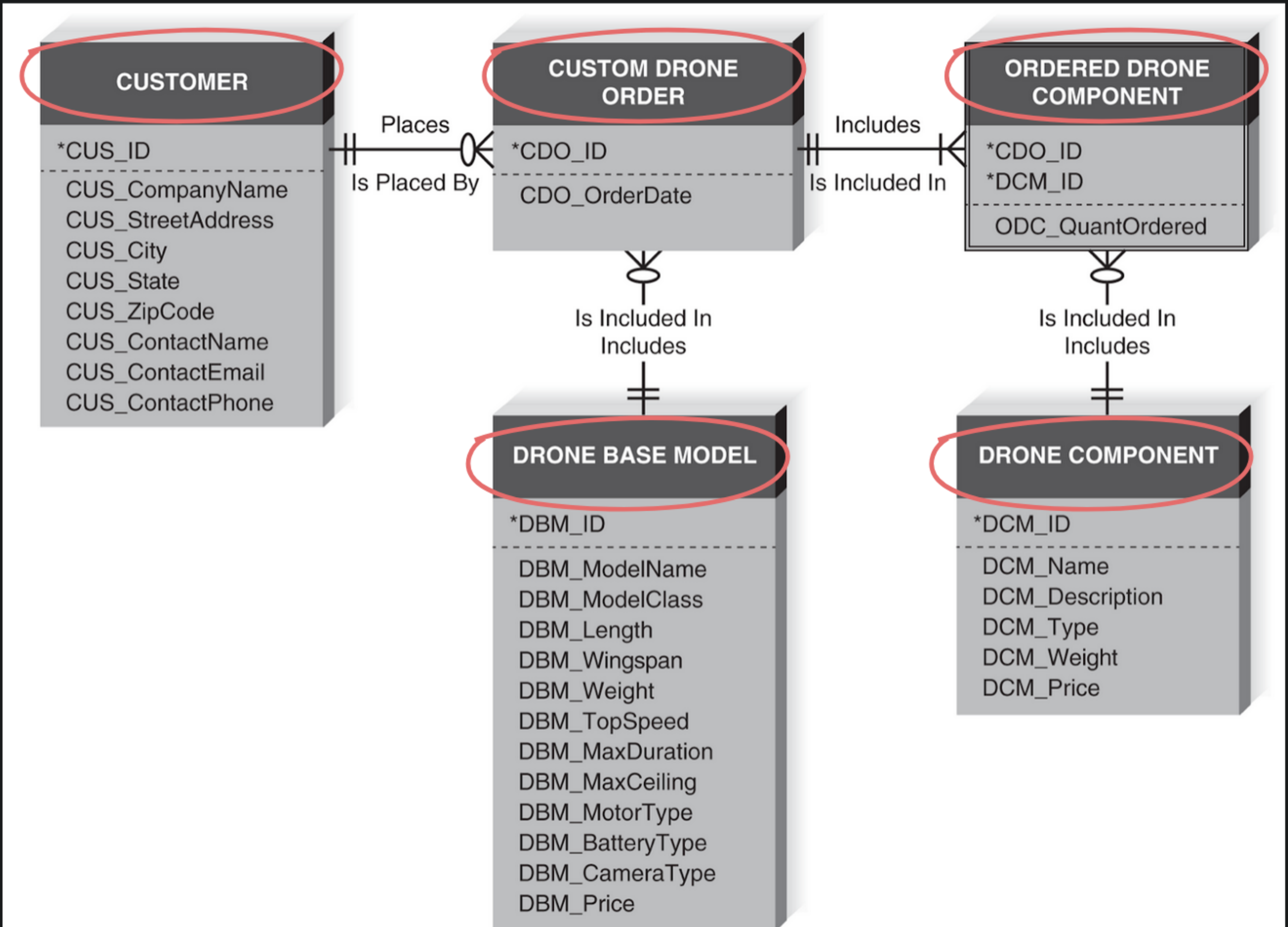


FIGURE 5-1 Partial DrönTeq sales system ERD.

- Five main categories (of data)
- A list of characteristics for each category

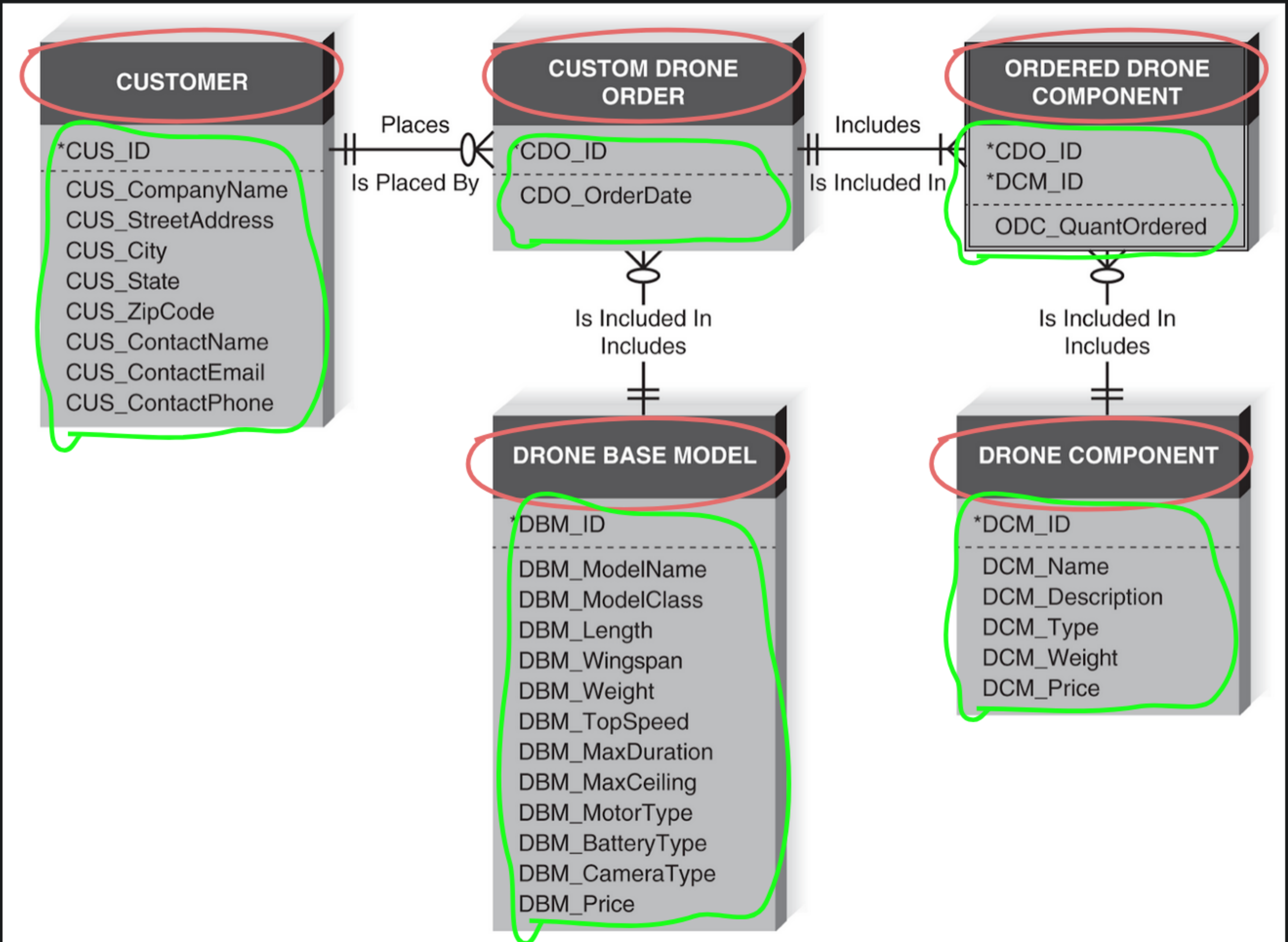


FIGURE 5-1 Partial DrönTeq sales system ERD.

- Five main categories (of data)
- A list of **characteristics** for each category
- An **identifier** we can use to uniquely identify each Customer, Order, Model, etc.

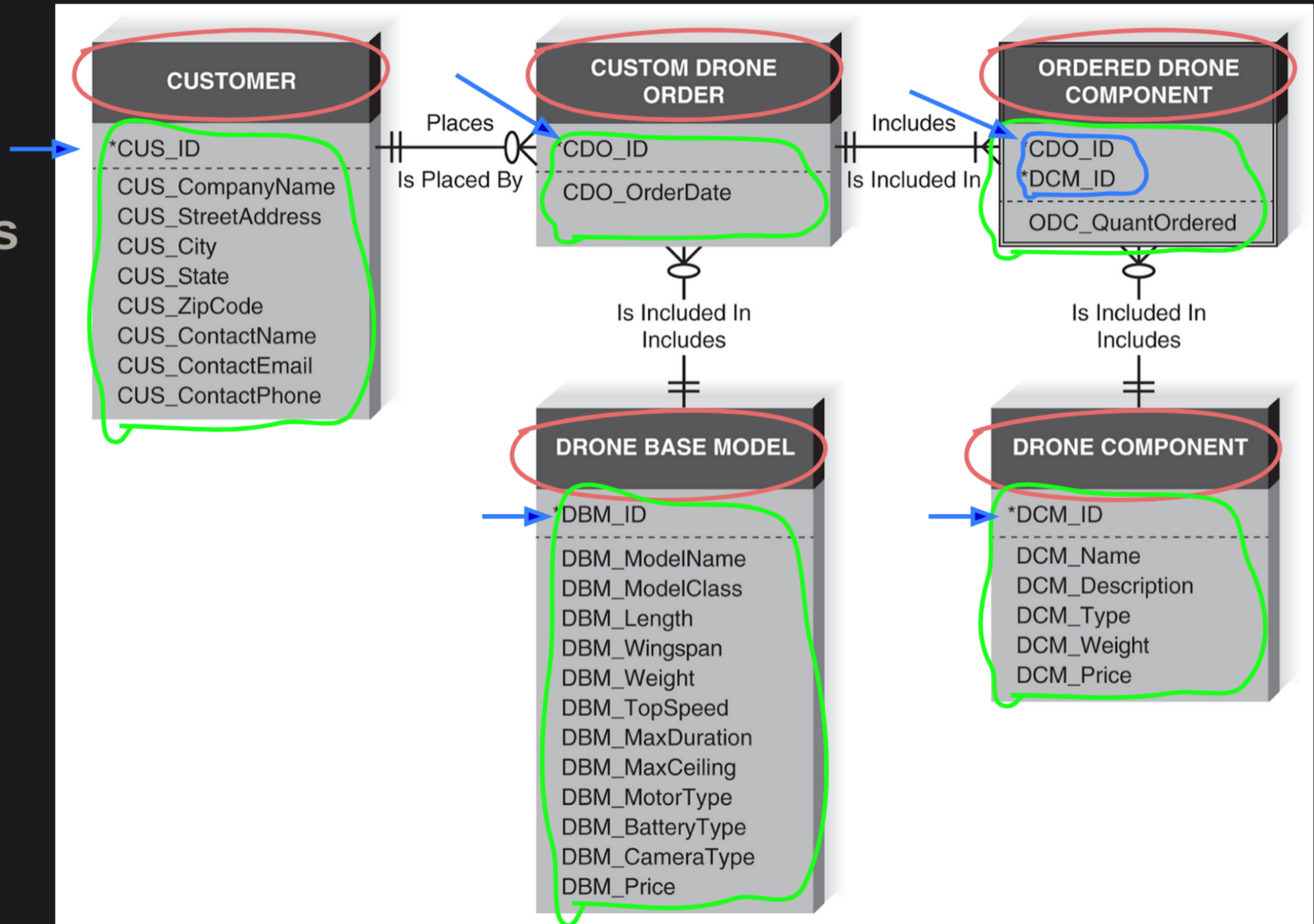


FIGURE 5-1 Partial DrönTeq sales system ERD.

- Five main categories (of data)
- A list of **characteristics** for each category
- An **identifier** we can use to uniquely identify each Customer, Order, Model, etc.
- The **relationships** between categories

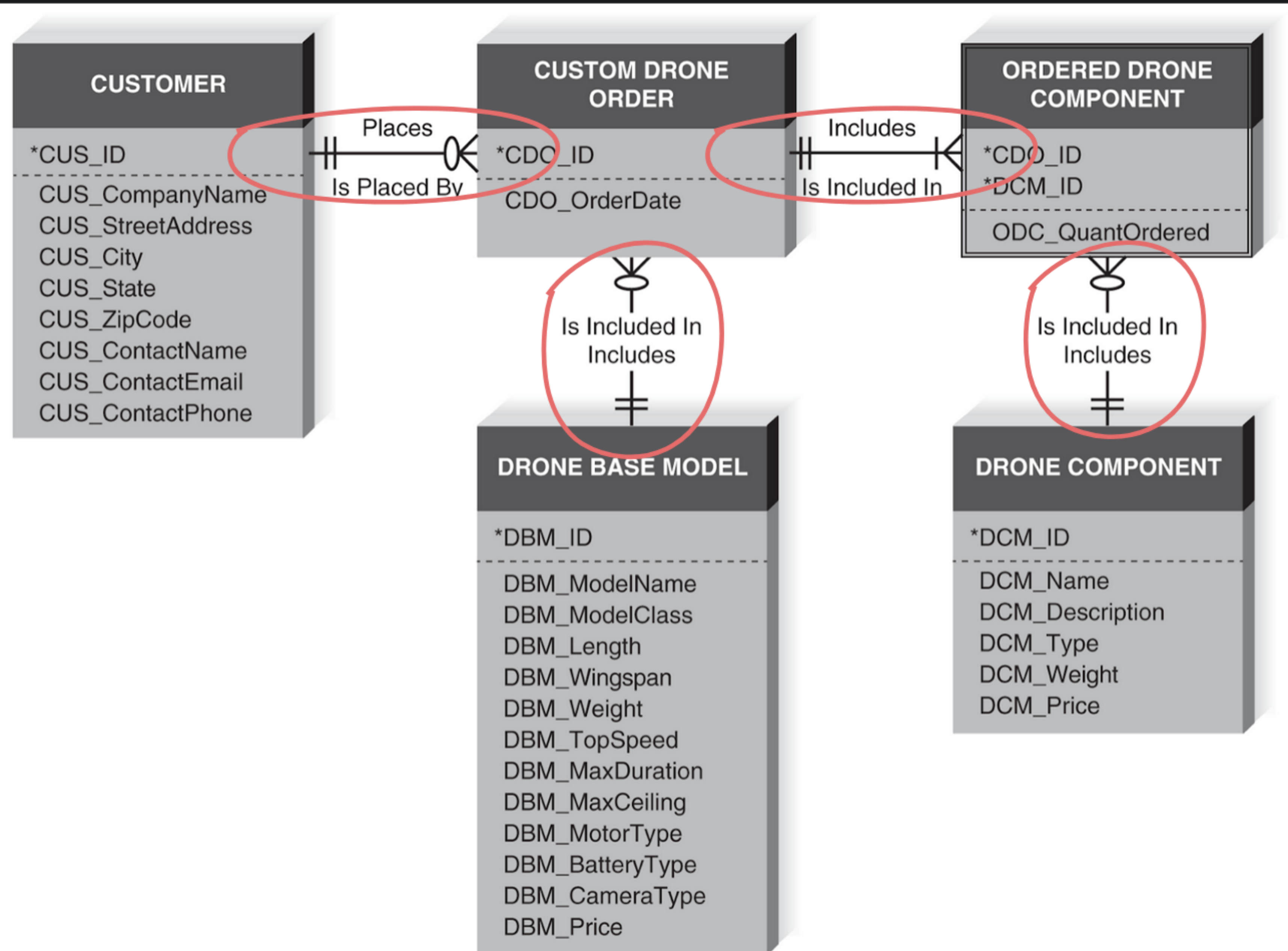


FIGURE 5-1 Partial DrönTeq sales system ERD.

- Five main categories (of data)
- A list of **characteristics** for each category
- An **identifier** we can use to uniquely identify each Customer, Order, Model, etc.
- The **relationships** between categories
- High-level **business rules**

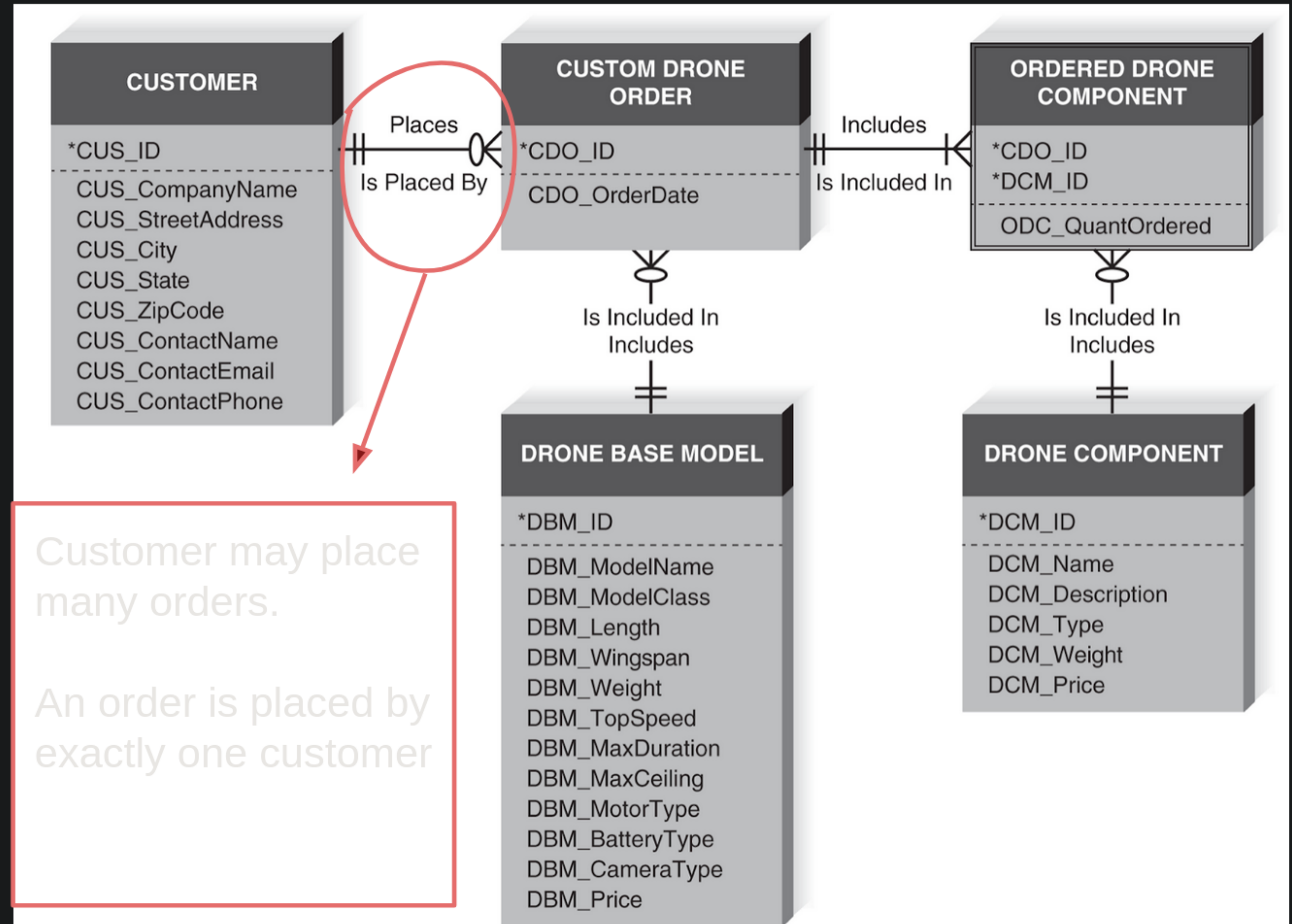


FIGURE 5-1 Partial DrönTeq sales system ERD.

Business rules are communicated by the relationships that the categories share and the **nature of those relationships.**

Elements of an Entity Relationship Diagram

A few notes:

- We use the **crow's foot notation**
- Entities represent something for which there exists multiple **instances** or occurrences.
 - If there is just one instance, it should not be considered an entity.
- Only the **attributes that are relevant** to the business process should be included.
- **Naming conventions** are important to minimize confusion.

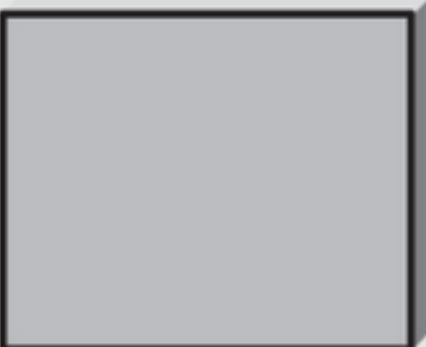
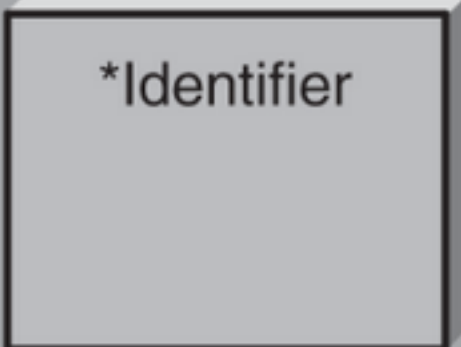
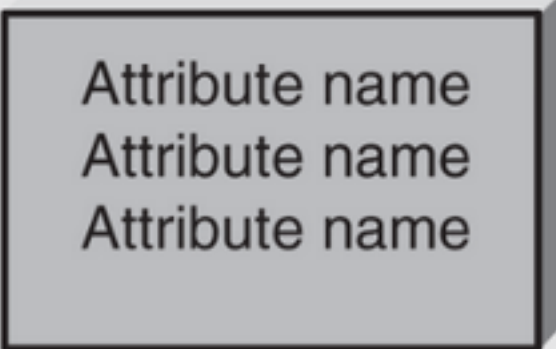
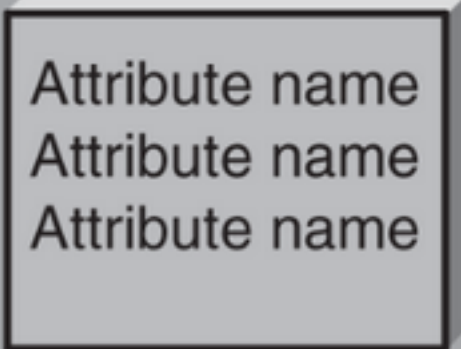
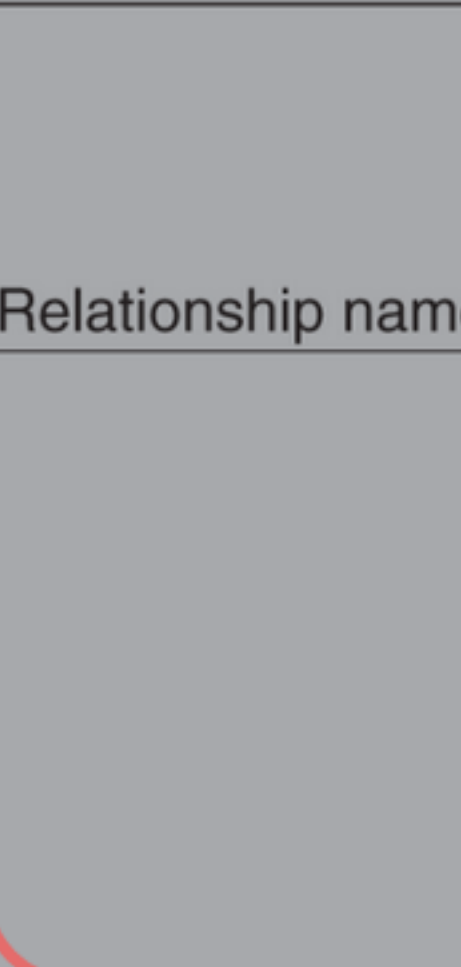
	IDEF1X	Chen	Crow's Foot
An ENTITY ✓ is a person, place, or thing. ✓ has a singular name spelled in all capital letters. ✓ has an identifier. ✓ should contain more than one instance of data.	ENTITY NAME 	ENTITY NAME 	ENTITY NAME 
An ATTRIBUTE ✓ is a property of an entity. ✓ should be used by at least one business process. ✓ is broken down to its most useful level of detail.	ENTITY NAME 		ENTITY NAME 
A RELATIONSHIP ✓ shows the association between two entities. ✓ has a parent entity and a child entity. ✓ is described with a verb phrase. ✓ has cardinality (1 : 1, 1 : N, or M : N). ✓ has modality (null, not null). ✓ is dependent or independent.	<u>Relationship name</u>		<u>Relationship name</u> 

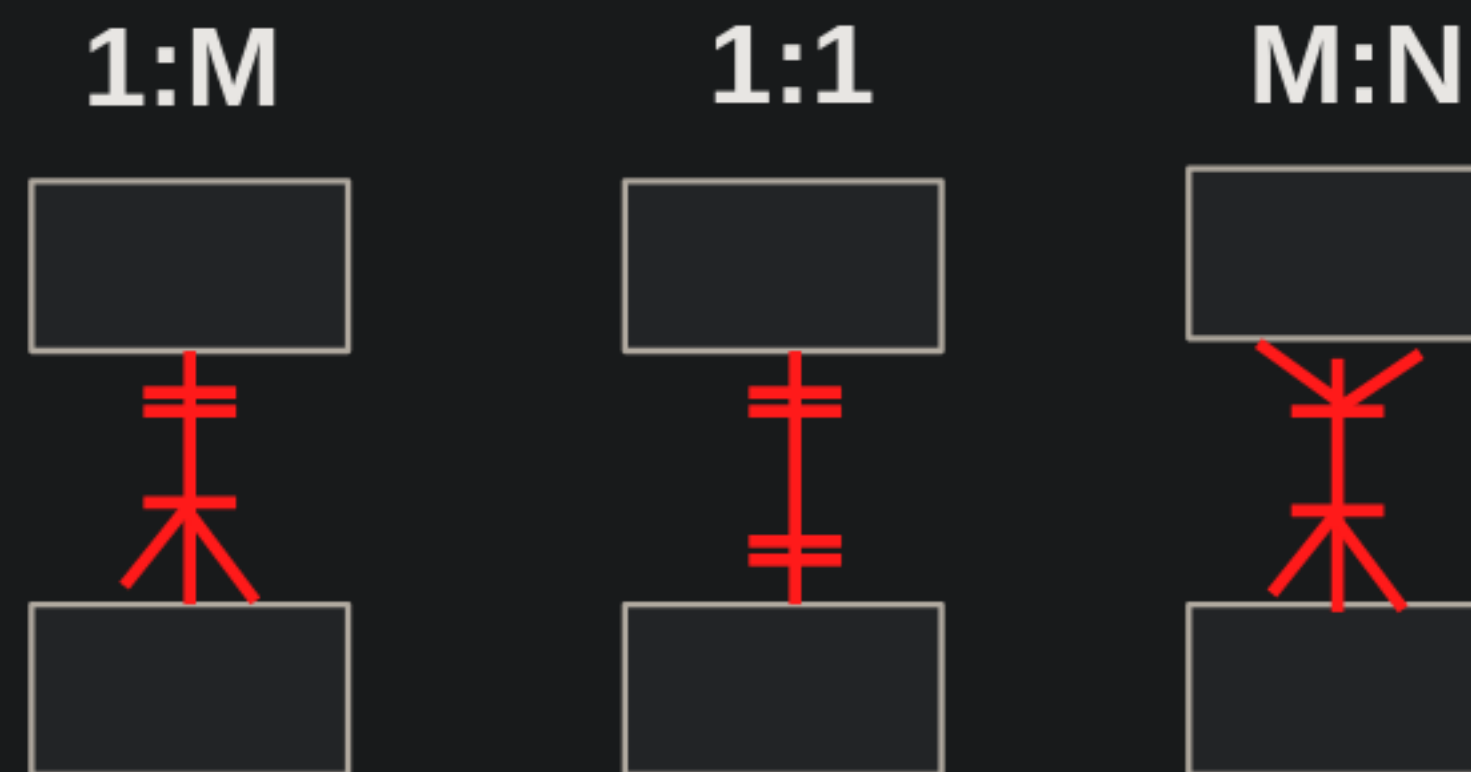
FIGURE 5-2 Data modeling symbol sets.

About Identifiers

- **One or more attributes** can serve as the identifier.
- When a combination of multiple fields is used as an identifier, we call it a **concatenated identifier**.

Cardinality

- Relationships are **bi-directional**.
- Cardinality is the ratio of **parent** instances to **child** instances.
- Ask **two questions to identify the cardinality**: *How many instances of one entity are associated with an instance of the other?*
 - Ask this question for each direction of the relationship



Cardinality

One to Many → 1:M

- A painter paints many paintings
- Each painting is painted by only one painter



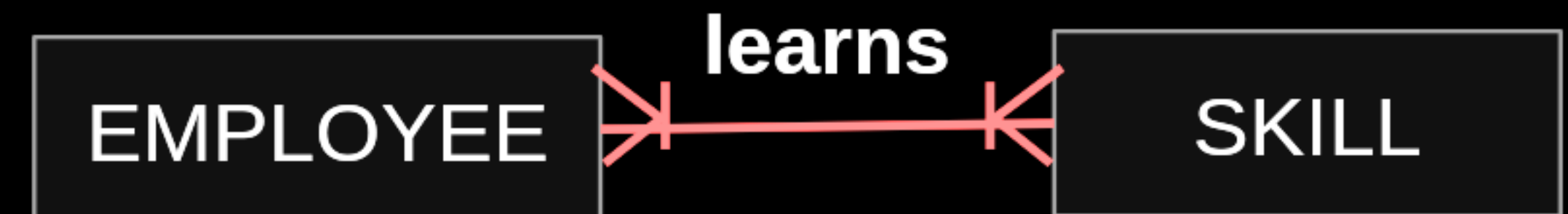
One to One → 1:1

- Each store is managed by a single employee
- Each store manager manages only a single store



Many to Many → M:N

- An employee may learn many job skills
- Each job skill may be learned by many employees



Modality

- Either **required** or **optional**
 - Whether an instance of an entity can exist without a related instance in the other entity?

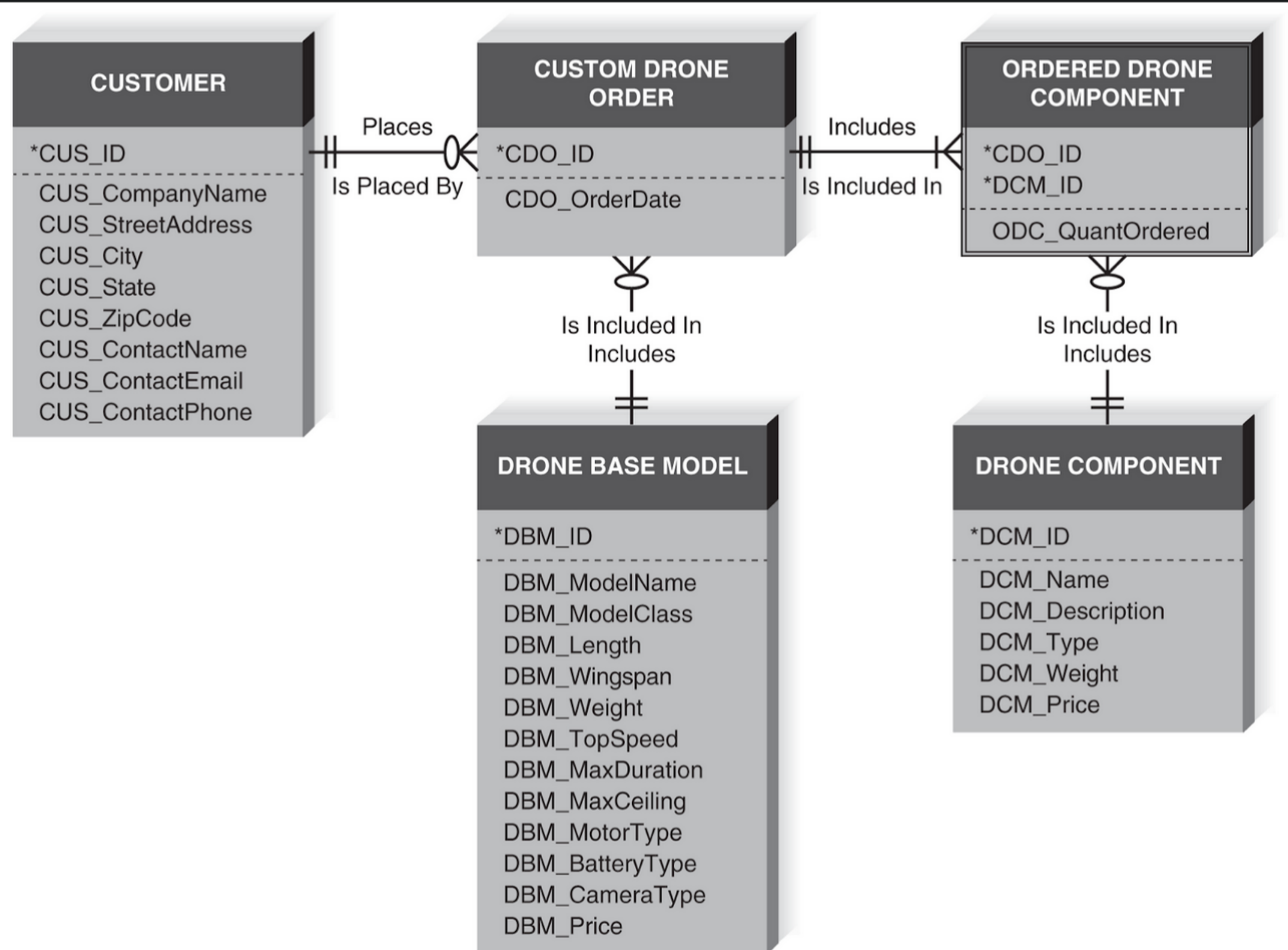


FIGURE 5-1 Partial DrönTeq sales system ERD.

□ Your Turn: Applying the Concepts of Cardinality and Modality

The Data Dictionary and Metadata

Metadata, is data about data.

Creating an Entity Relationship Diagram

Peer Review Data

I usually assign students to perform two **peer reviews** during the term if they are working on a group project together. Here's an example of the data that is generated as students fill out the peer review form:

1	Timestamp	What is your name?	Who are you reviewing?	Presence	Contributions	Team Skills	Communication	Group
3	04/03/2020 10:34:57	Fraser	Fraser	Excellent	Acceptable	Acceptable	Poor	Set B - Group 4
4	04/03/2020 10:35:12	Mattias	Mattias	Excellent	Excellent	Excellent	Excellent	Set A - Group 2
5	04/03/2020 10:35:30	Kevin	Kevin	Excellent	Acceptable	Excellent	Excellent	Set B - Group 3
6	04/03/2020 10:35:30	Jeremy	Fraser	Excellent	Poor	Acceptable	Acceptable	Set B - Group 4
7	04/03/2020 10:35:34	Jakob	Jakob	Excellent	Excellent	Acceptable	Excellent	Set B - Group 1
8	04/03/2020 10:35:36	Kevin	Kevin	Excellent	Excellent	Excellent	Excellent	Set B- Group 5
9	04/03/2020 10:35:42	Ben	Ben	Excellent	Acceptable	Acceptable	Acceptable	Set B - Group 3
10	04/03/2020 10:35:43	Mattias	Vicly	Excellent	Excellent	Excellent	Excellent	Set B - Group 2
11	04/03/2020 10:35:53	Dylan	Dylan	Excellent	Excellent	Excellent	Excellent	Set B - Group 3
12	04/03/2020 10:35:54	Jonathan	Myself (Jonathan)	Acceptable	Acceptable	Excellent	Excellent	Set B - Group 1
13	04/03/2020 10:36:02	Prerna	myself	Excellent	Excellent	Excellent	Acceptable	Set B - Group 2
14	04/03/2020 10:36:09	Kevin	Oves	Excellent	Excellent	Excellent	Excellent	Set B- Group 5
15	04/03/2020 10:36:12	Jeremy	Kevin	Acceptable	Excellent	Excellent	Excellent	Set B - Group 4
16	04/03/2020 10:36:15	Dylan	Ben	Excellent	Excellent	Excellent	Excellent	Set B - Group 3
17	04/03/2020 10:36:21	Fraser	Manreet	Excellent	Acceptable	Acceptable	Acceptable	Set B - Group 4

Peer Review Data

1	Timestamp	What is your name?	Who are you reviewing?	Presence	Contributions	Team Skills	Communication	Group
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17	04/03/2020 10:36:21	Fraser	Manreet	Excellent	Acceptable	Acceptable	Acceptable	Set B - Group 4

I'm going to walk you through **the process** for designing a data model.

The process of data modeling

1. Detect **entities** and the **relationships** between them.
2. Specify the **cardinality** of relationships (1:1, 1:M or M:N).
3. Specify the **modality** of each relationship to each entity.
4. Break down M:N relationships into two 1:M relationships using an **intersection entity**.
5. Add **attributes** and assign **identifiers**



Jimmy Kong

Oct 20, 2023

In textbook and practice quiz you gave us, adding attributes and identifiers is the second step; While on the slide it is the last step. May I ask which one is the correct answer?



Maryam Khezzadeh

Oct 21, 2023

Yes there is a difference between the order of steps here and in the textbook. Frankly, it doesn't matter.

[Show more](#)

[2 more replies](#)



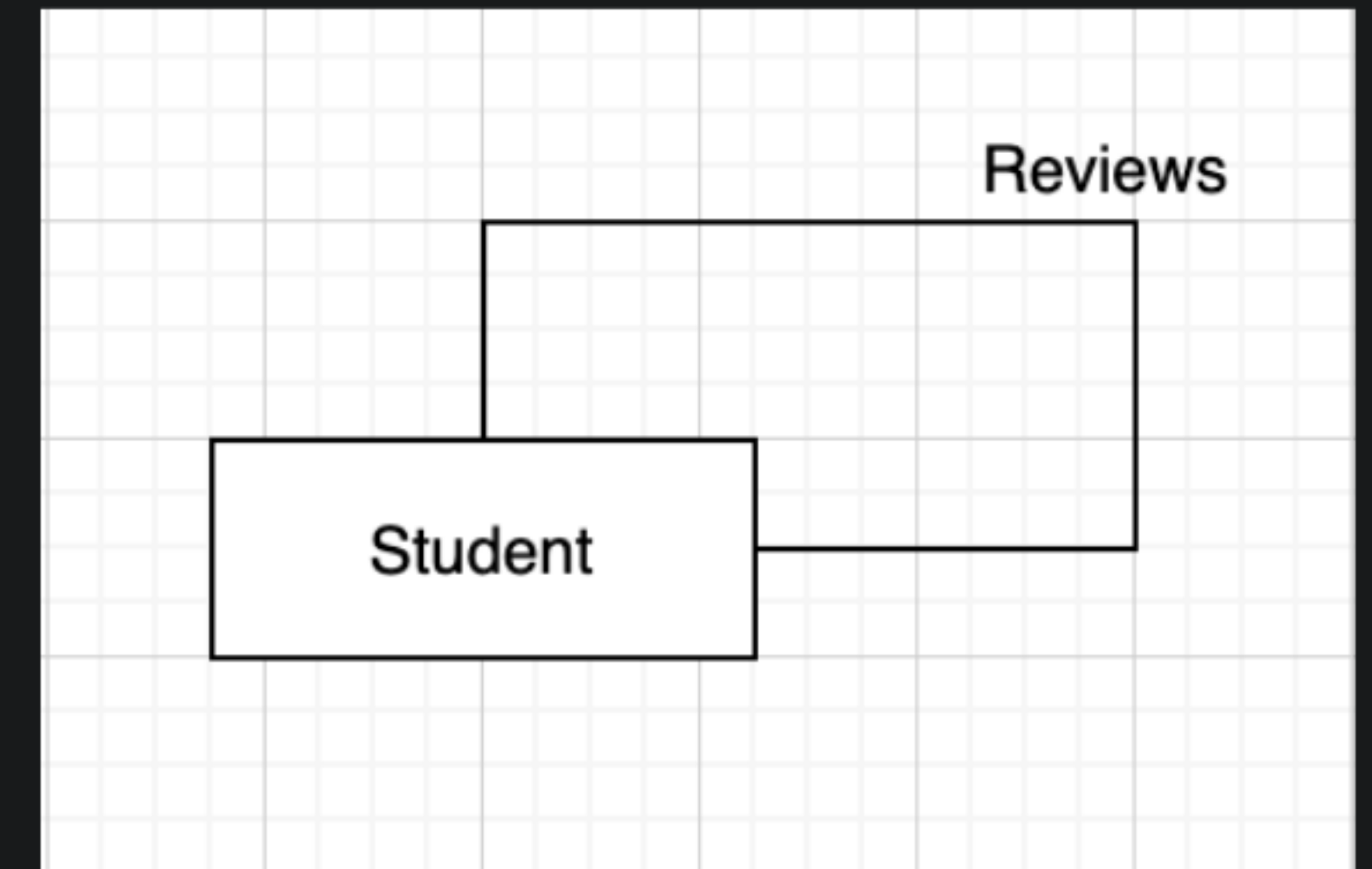
Maryam Khezzadeh

Feb 20, 2024

Re-opened

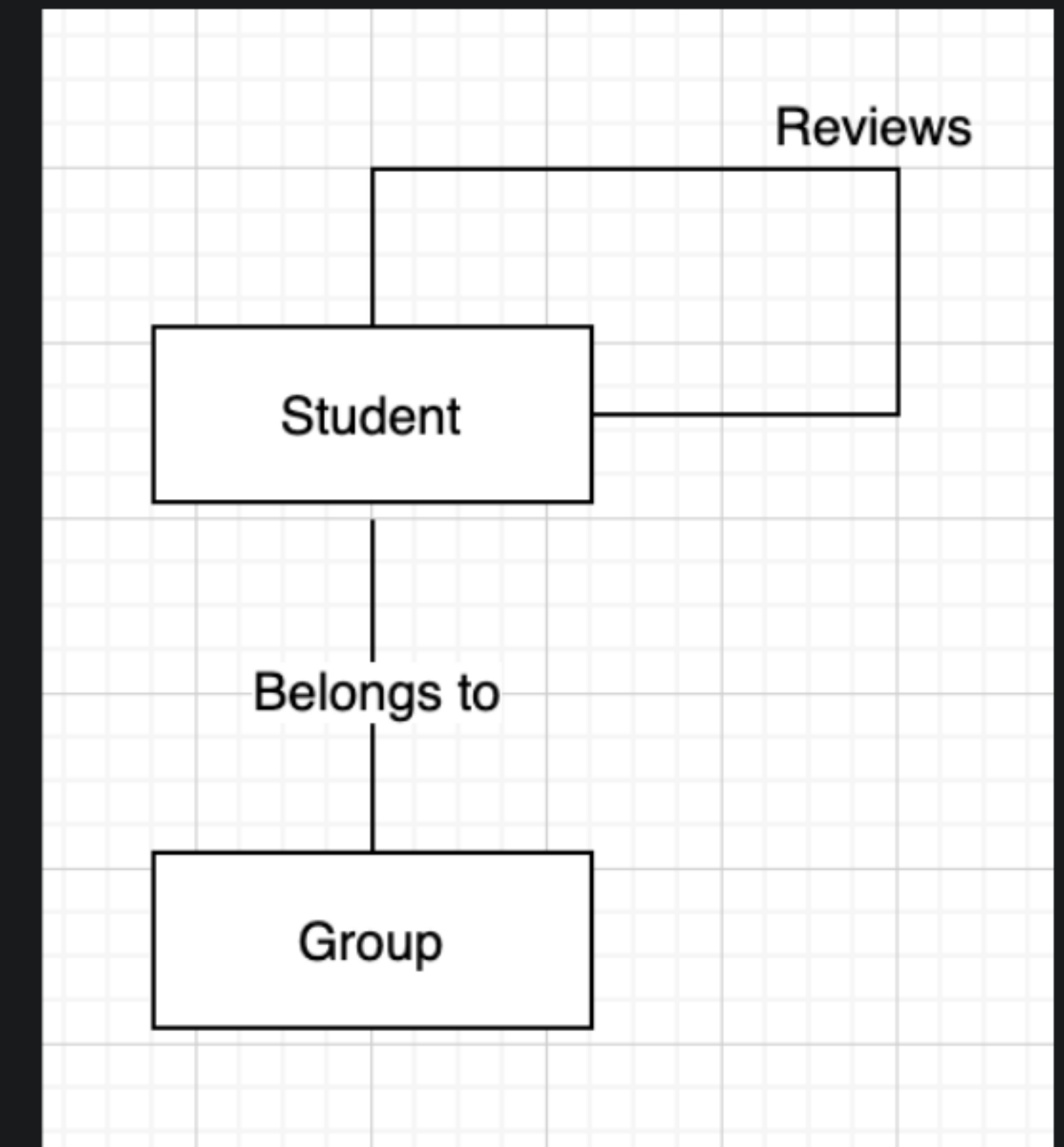
Step 1: Entities & Relationships

- A student reviews another student.



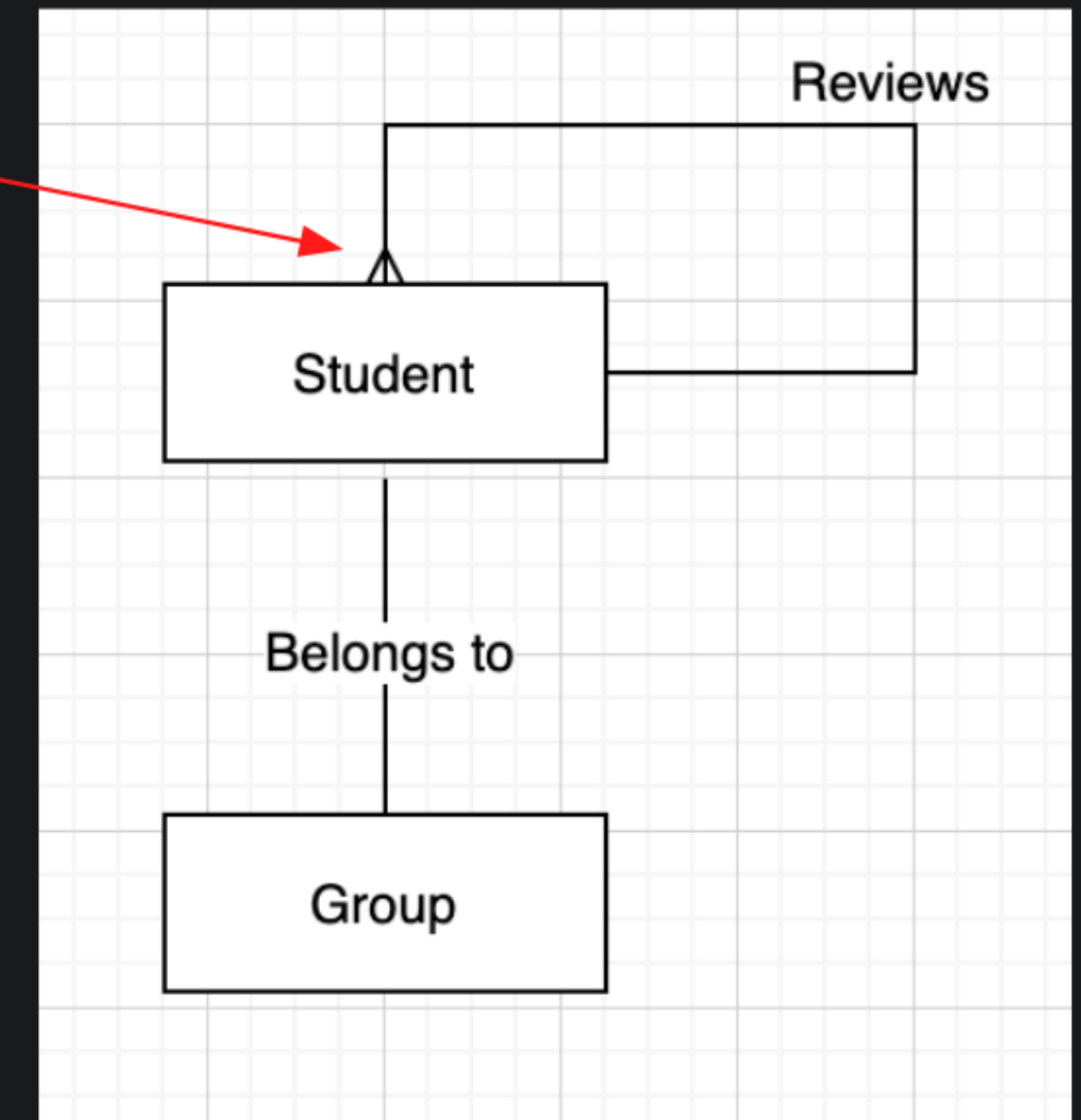
Step 1: Entities & Relationships

- A student reviews another student.
- Each student can only review members of their own group.



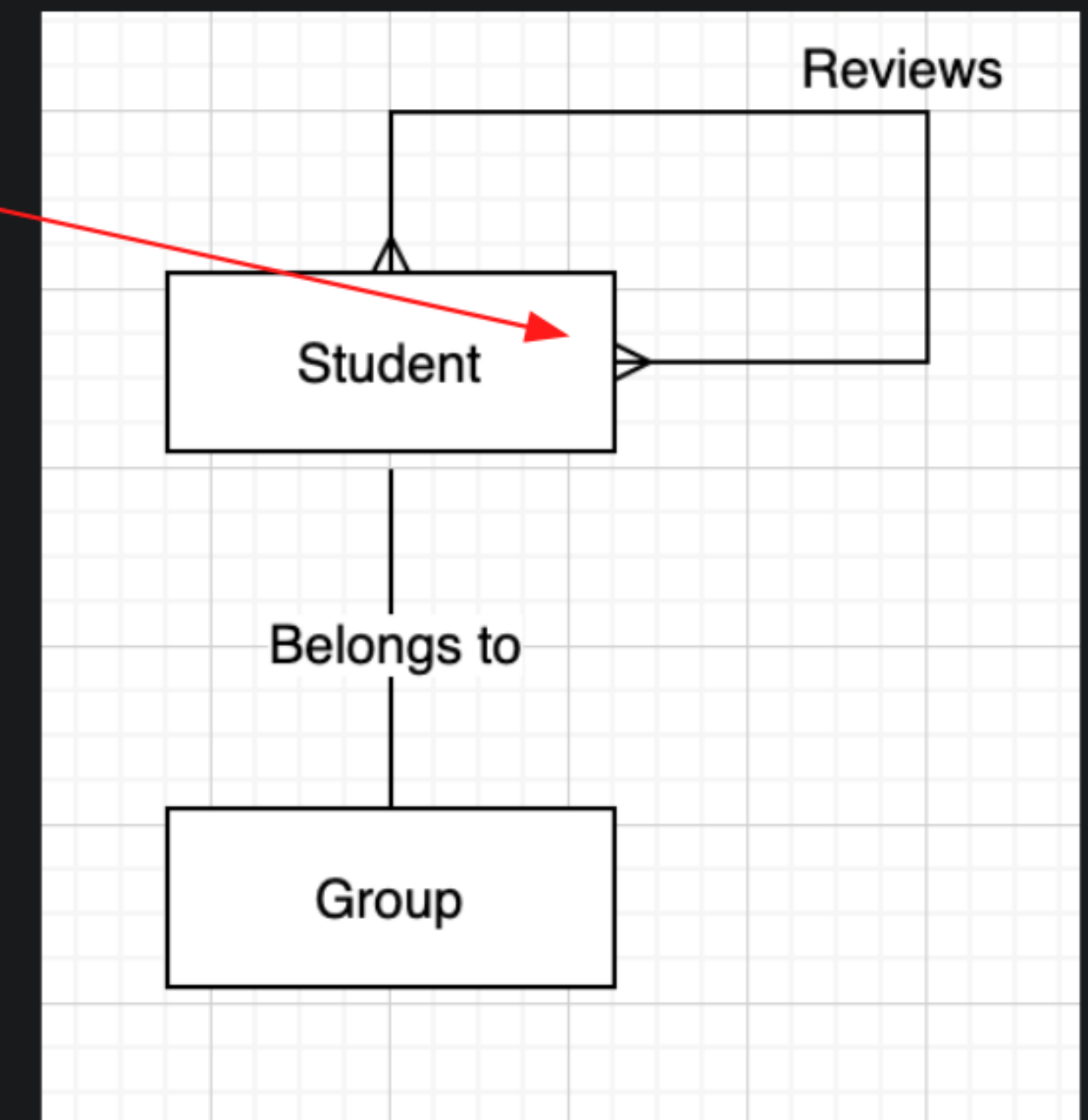
Step 2: Relationship Cardinality

- A student reviews another student.
- Each student can only review members of their own group.
- Each student can review multiple other students.



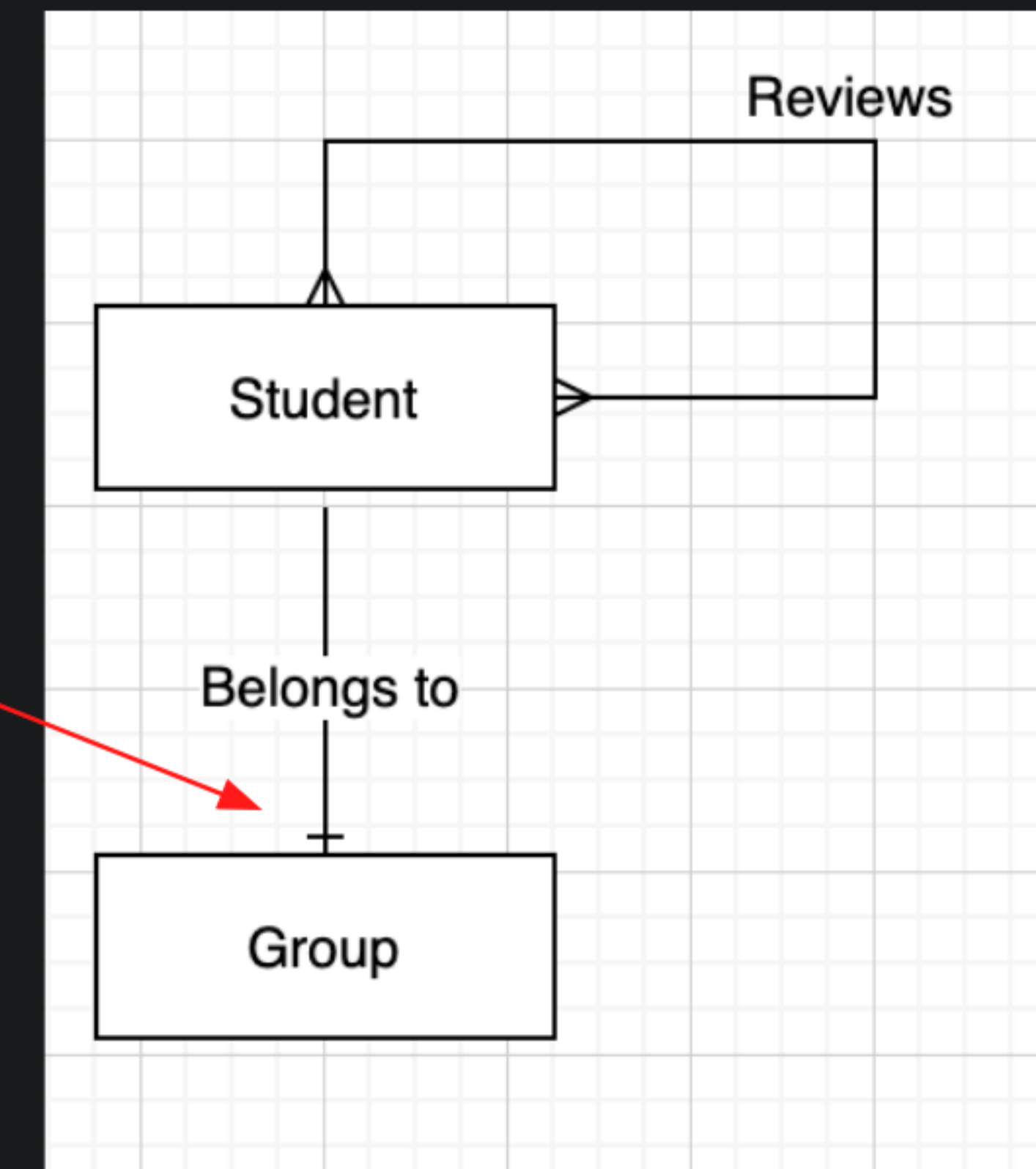
Step 2: Relationship Cardinality

- A student reviews another student.
- Each student can only review members of their own group.
- Each student can review multiple other students.
- Each student, in turn, is reviewed multiple times themselves.



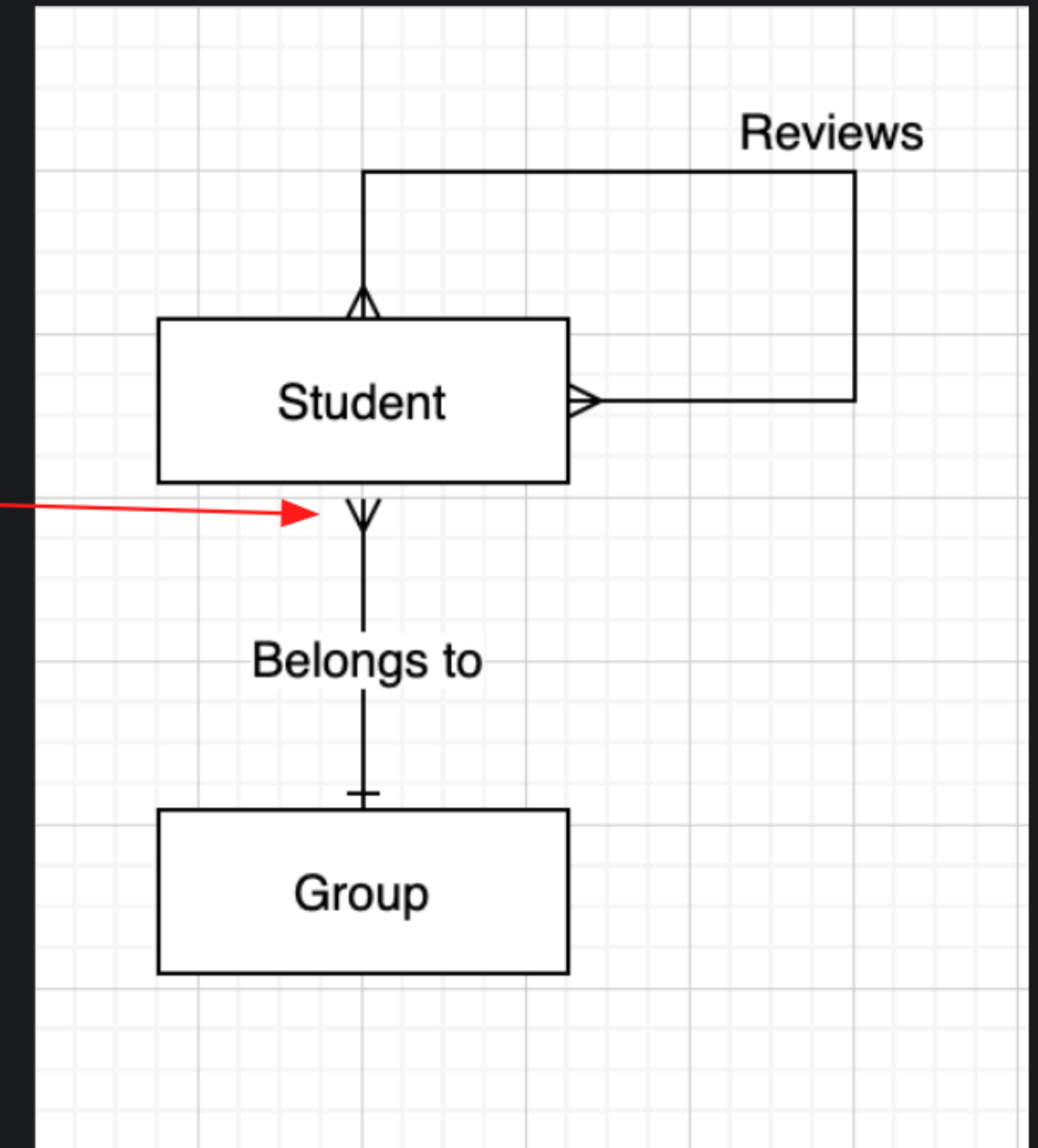
Step 2: Relationship Cardinality

- A student reviews another student.
- Each student can only review members of their own group.
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- Each student, in turn, is reviewed multiple times themselves.
- Every student belongs to only one group.



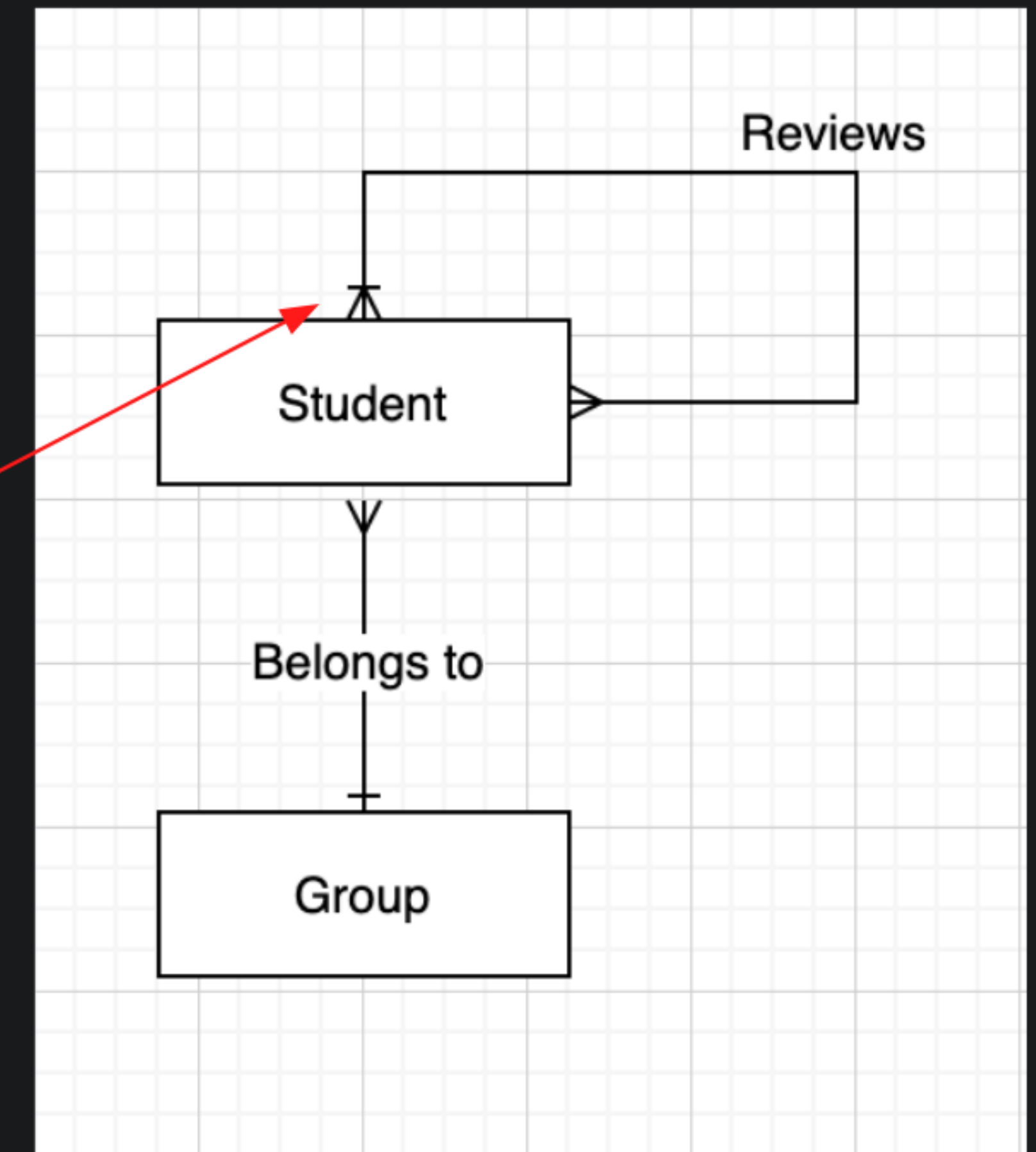
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- Each student, in turn, is reviewed multiple times themselves.
- Every student belongs to only one group.
- Each group can have multiple members.



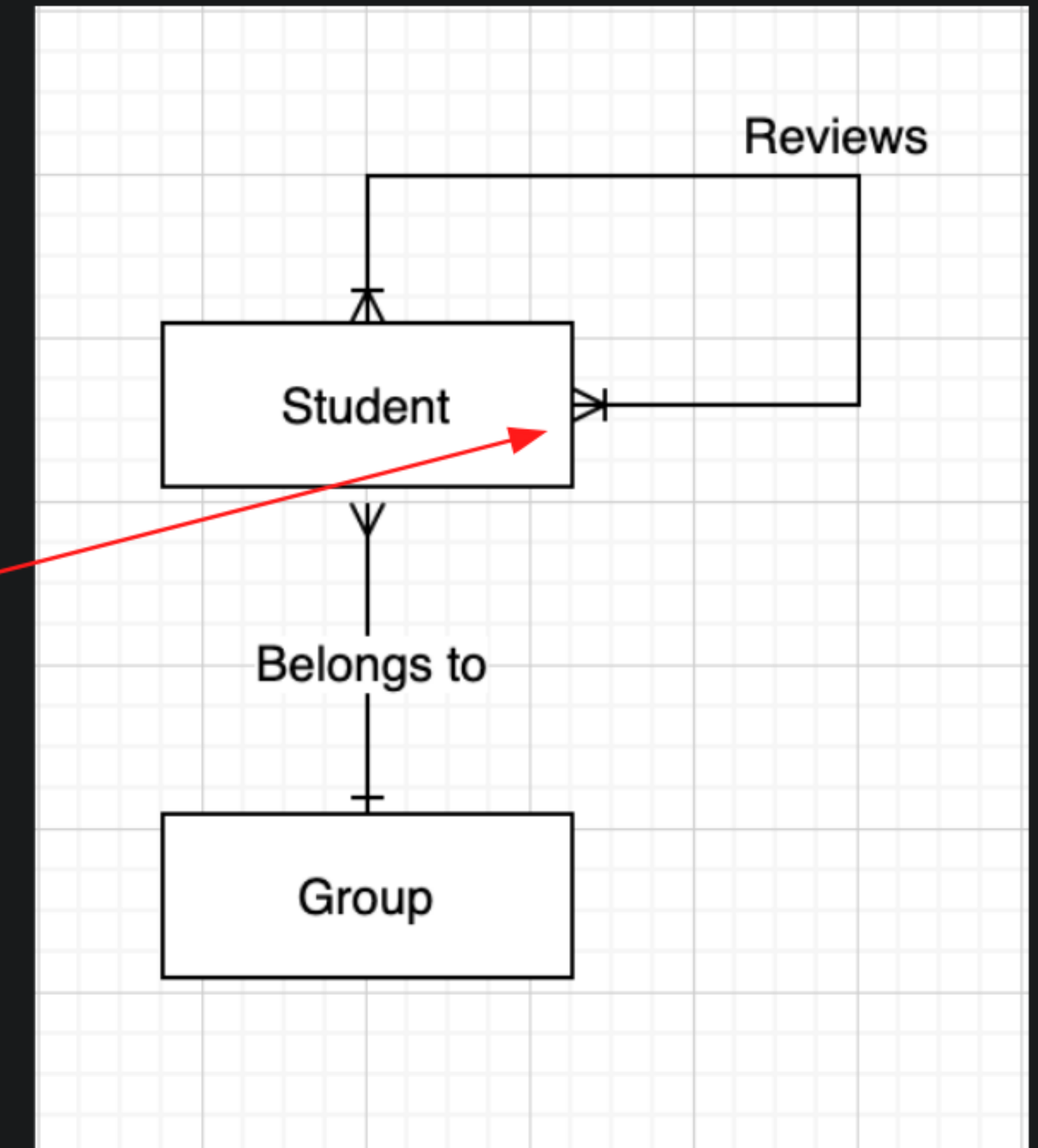
Step 3: Relationship Modality

- A student reviews another student.
- Each student can only review members of their own group.
- Each student can review multiple other students.
- Each student, in turn, is reviewed multiple times themselves.
- Every student belongs to only one group.
- Each group can have multiple members.
- A student must review all members of their group (including themselves)



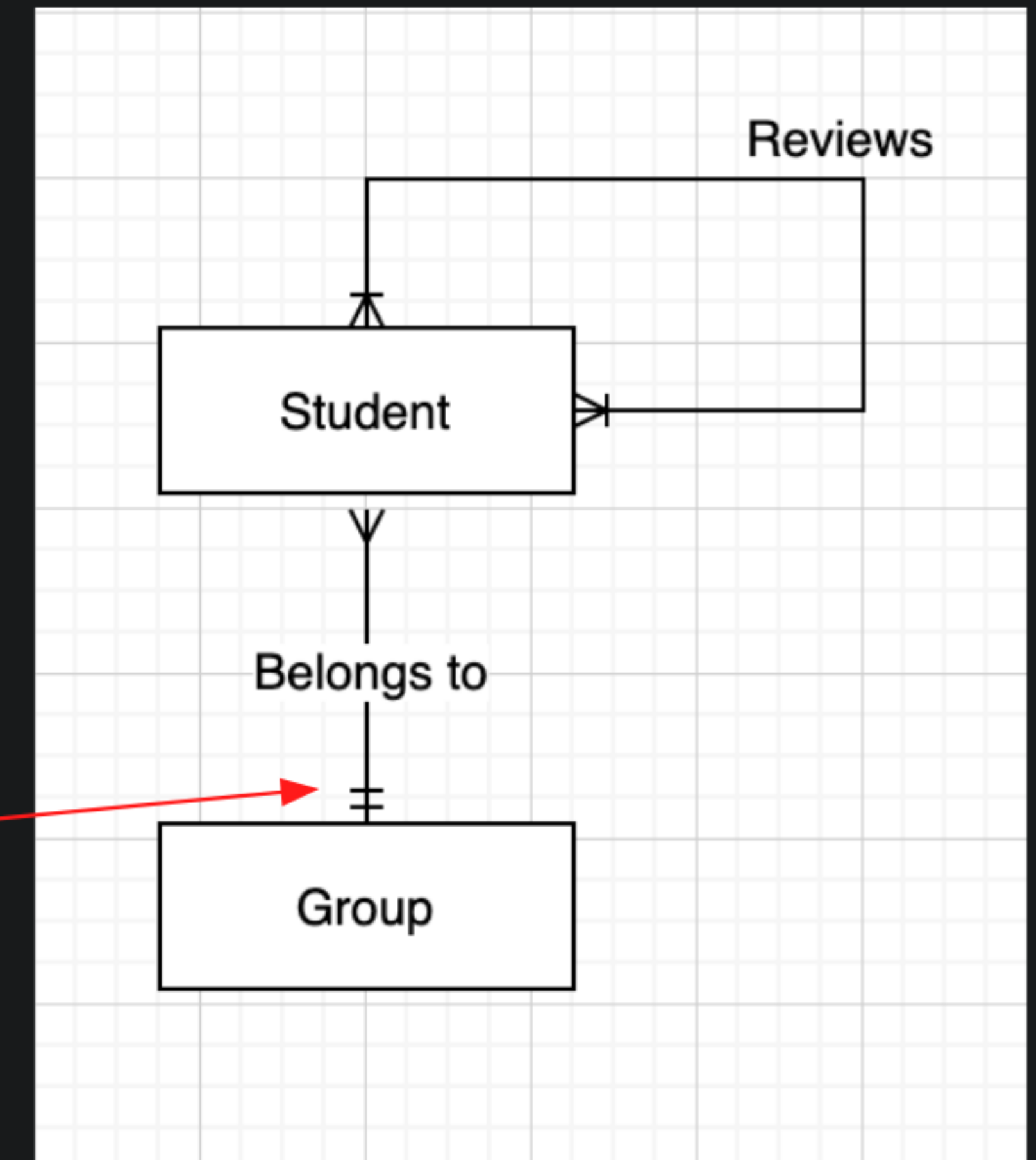
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- Each student, in turn, is reviewed multiple times themselves.
- Every student belongs to only one group.
- Each group can have multiple members.
- A student must review all members of their group (including themselves)
- Each student receives at least 1 review.



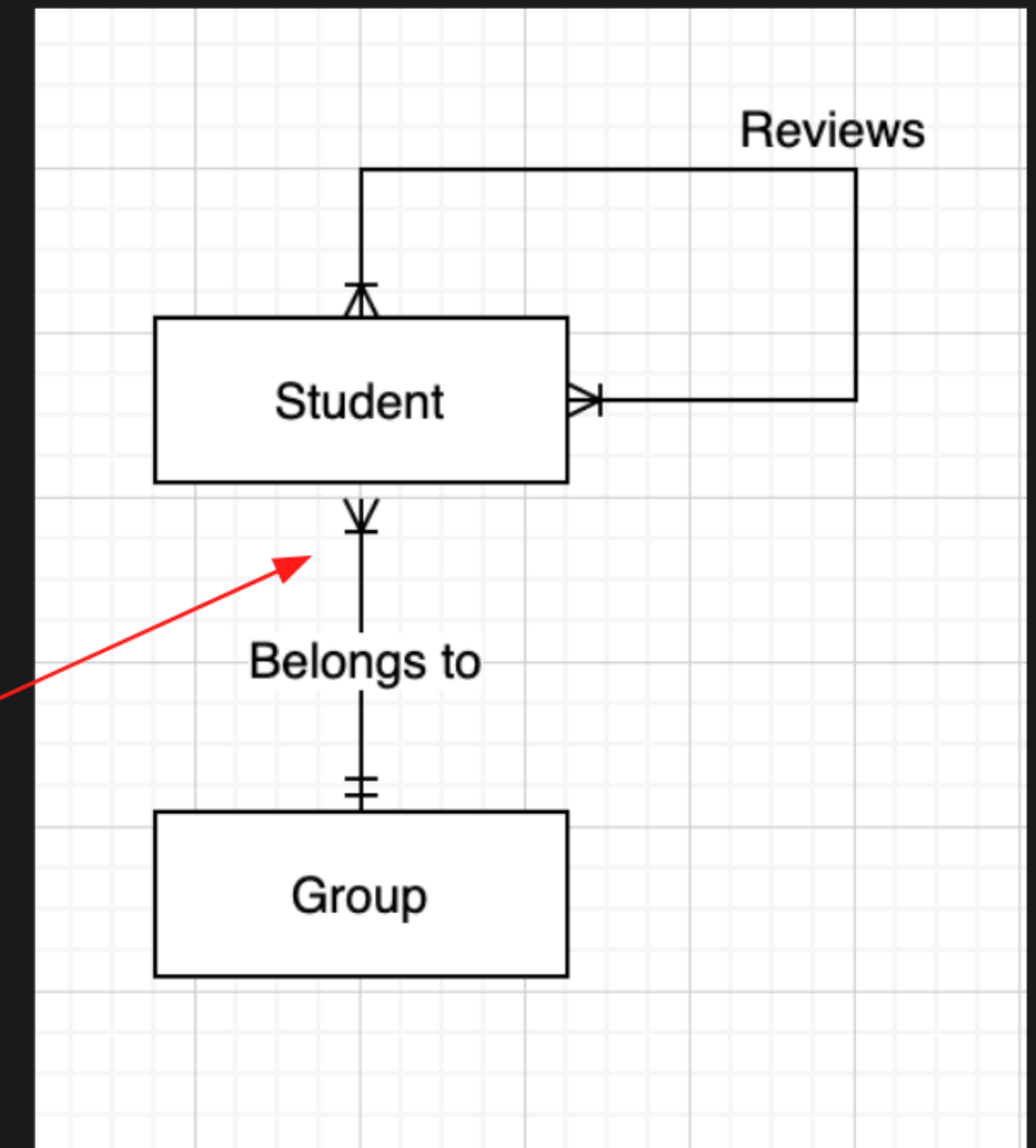
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- Each student receives at least 1 review.
- Each student must belong to a group.



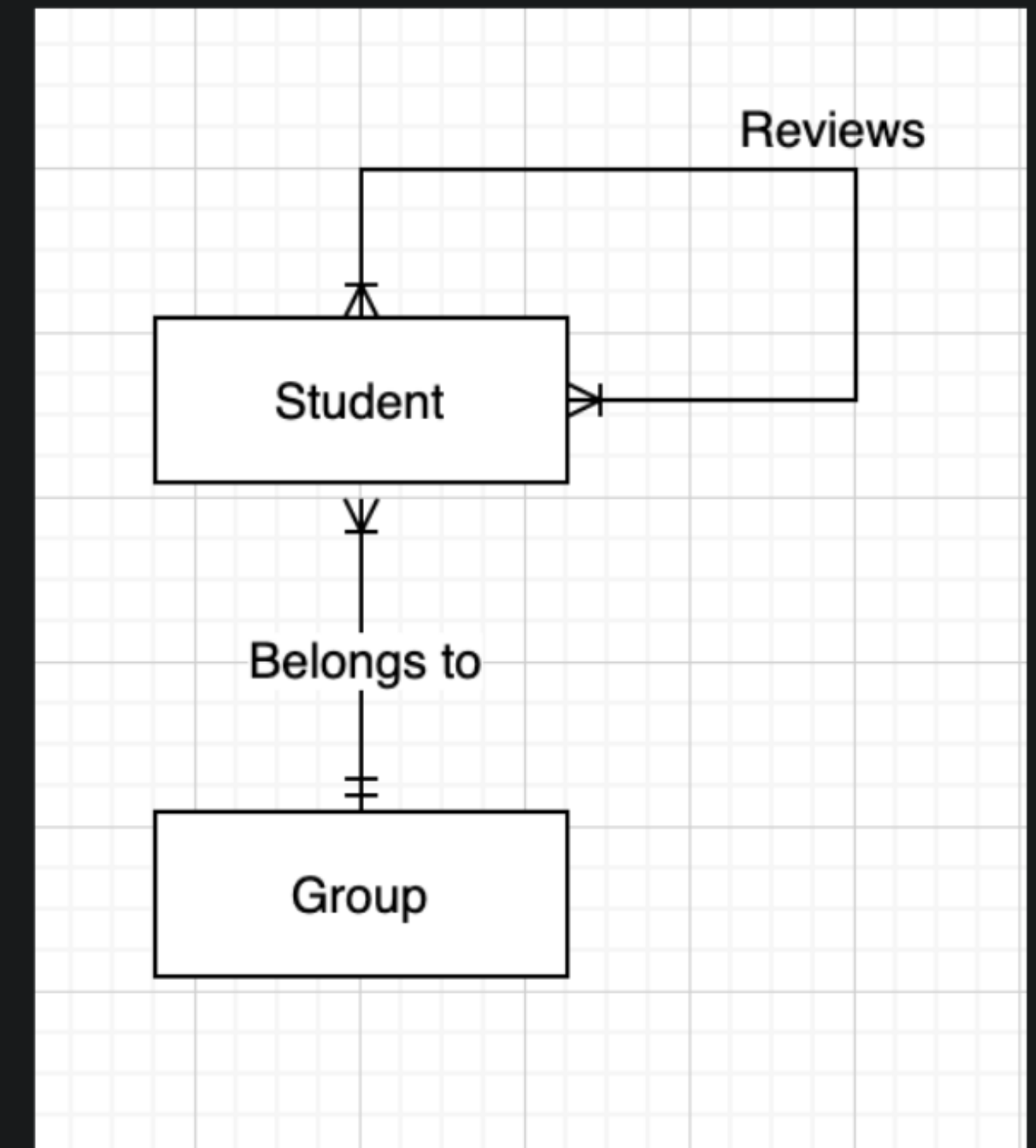
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- Each group can have multiple members.
- A student must review all members of their group (including themselves)
- Each student receives at least 1 review.
- Each student must belong to a group.
- Each Group must have at least one member.



Step 3: Relationship Modality

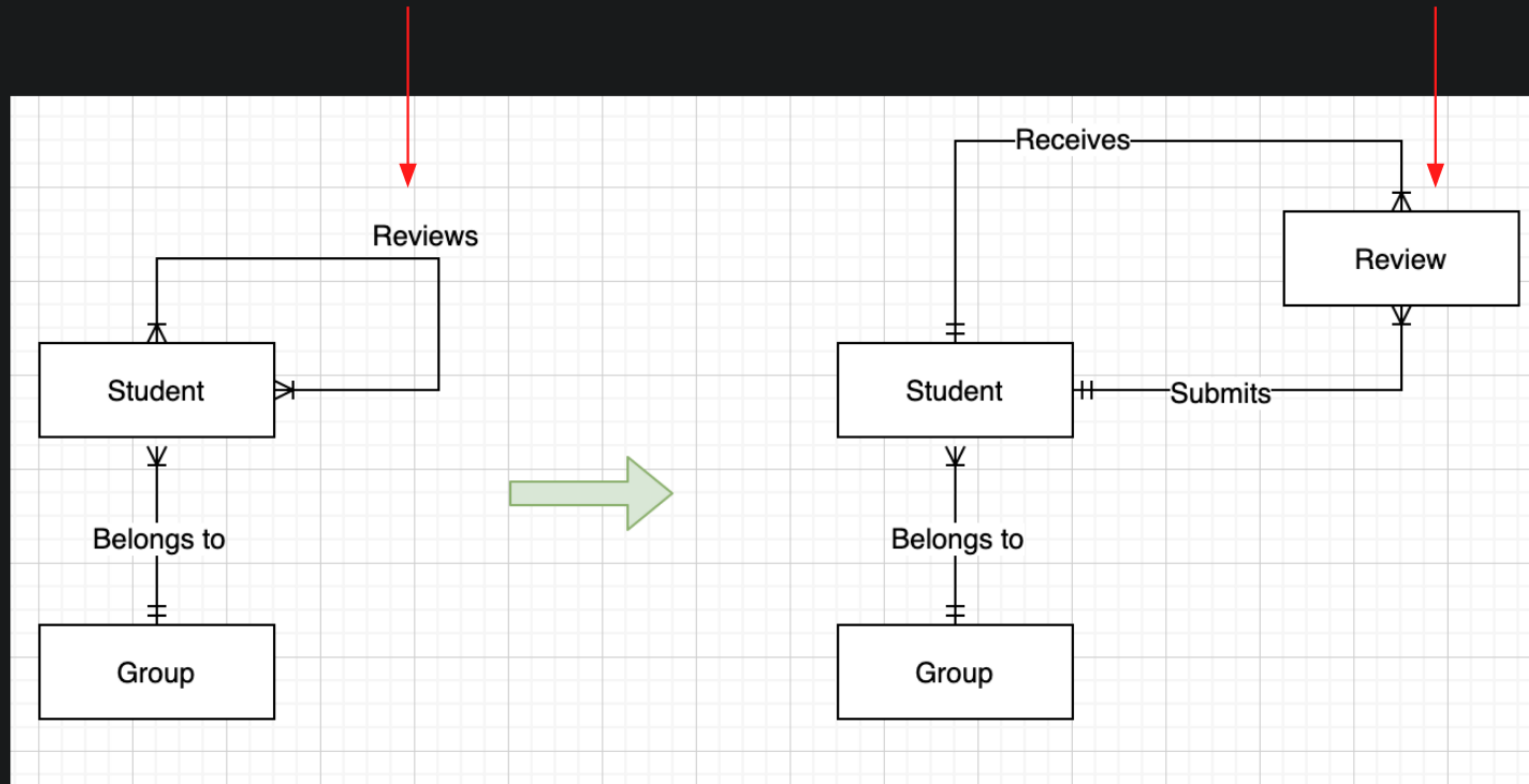
- A student reviews another student.
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- Each student can review multiple other students.
- Each student, in turn, is reviewed multiple times themselves.
- Every student belongs to only one group.
- Each group can have multiple members.
- A student must review all members of their group (including themselves)
- Each student receives at least 1 review.
- Each student must belong to a group.
- Each Group must have at least one member.



Business Rules: Simple descriptive sentences that help us establish and validate entities and relationships

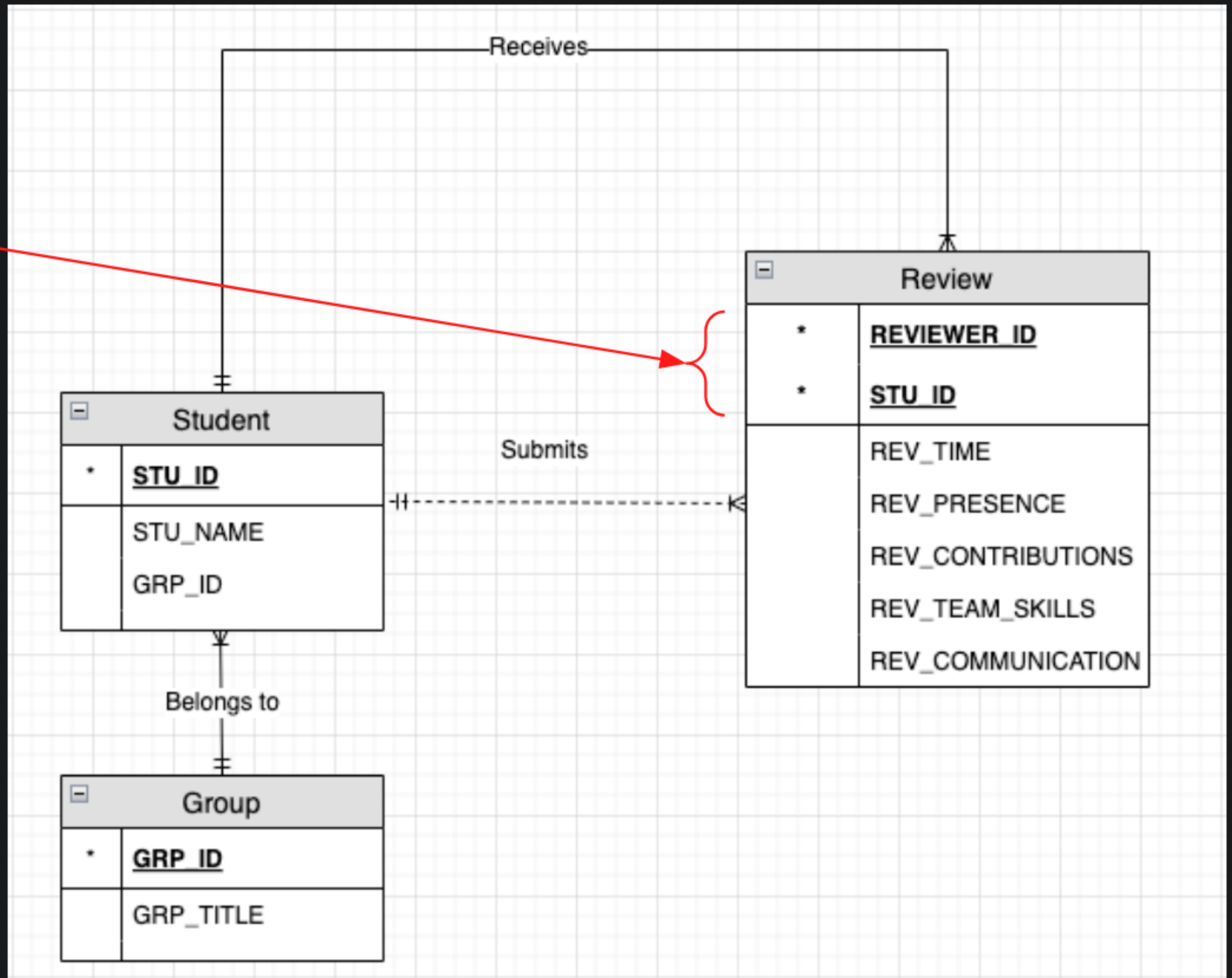
Step 4: Intersection Entities

Break down **M:N relationships** into two 1:M relationships using an **Intersection entity**.



Step 5: Attributes & Identifiers

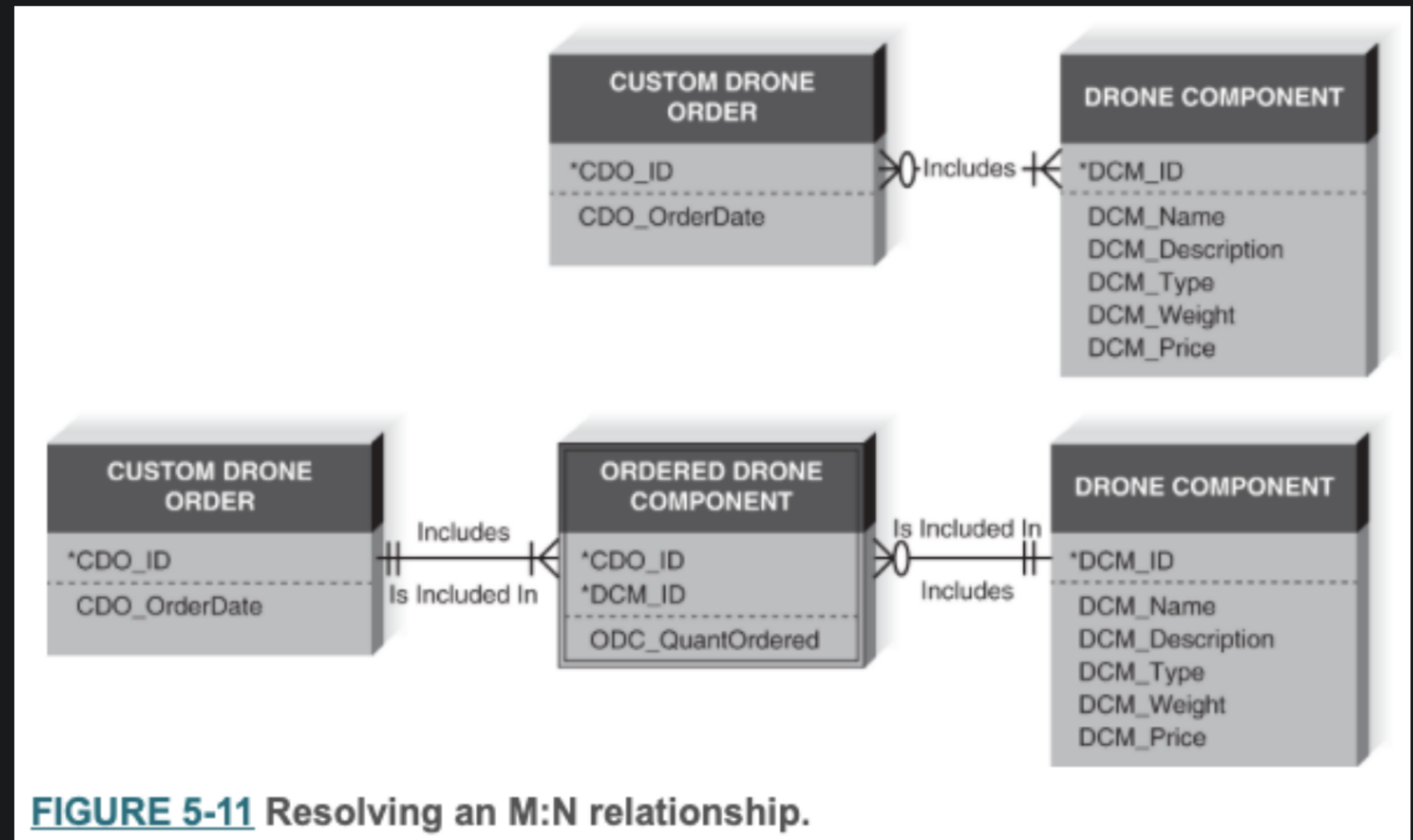
intersection entities often have a composite Identifier made of the Identifiers of the two entities they associate.



Advanced Syntax

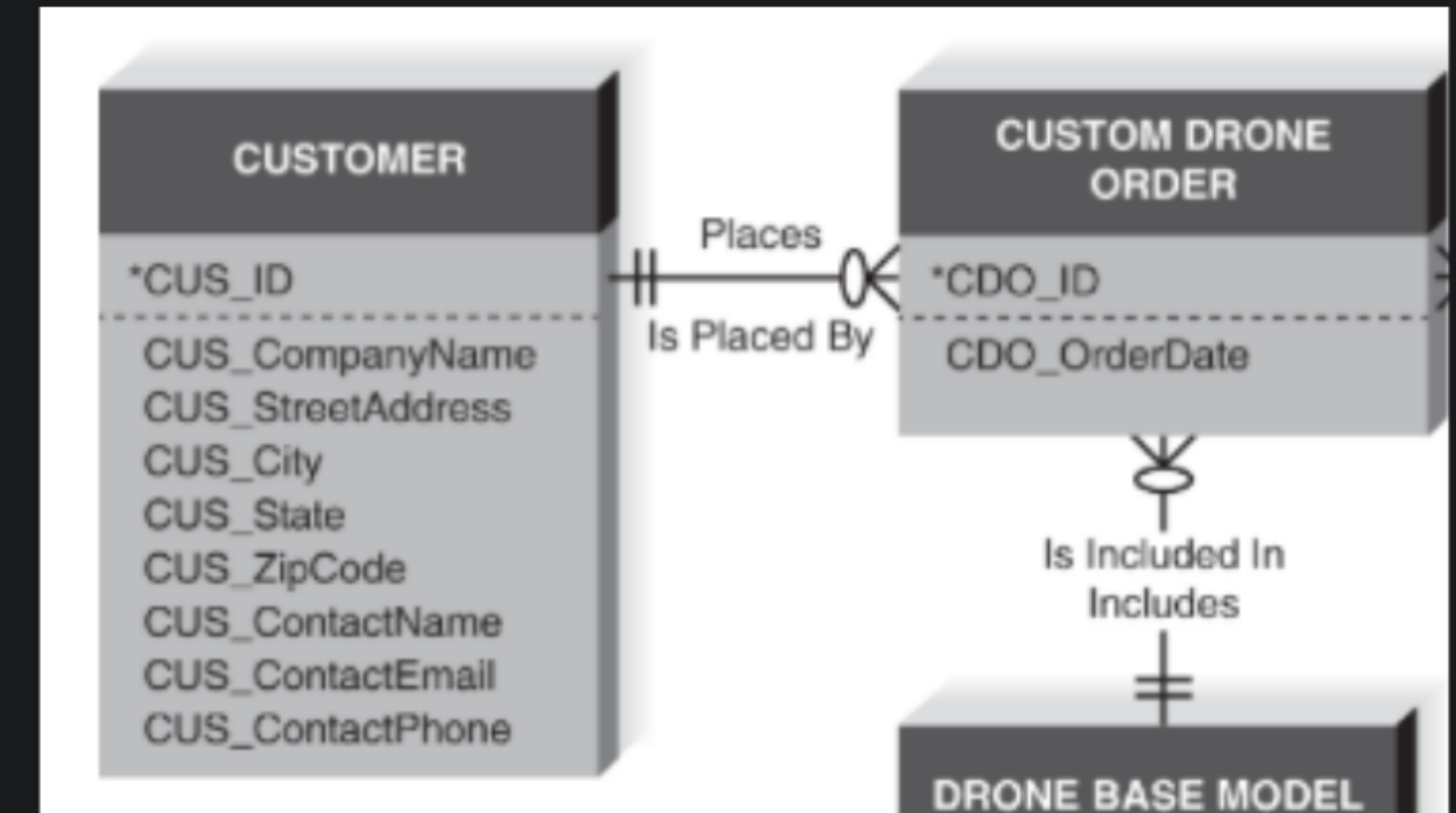
Intersection Entity

- Exists to capture some **information about the (M:N) relationship** between two other entities.
- Also called **associative entities**.
- We resolve a M:N relationship into two 1:M relationships.



Independent Entity

- Can exist without the help of another entity
 - E.g. Customer, Base Model, Component
 - They have **identifiers created from their own attributes**.
 - I.e. Attributes from other entities were not needed to uniquely identify instances of these entities.
- Relationships that include independent entities are called **non-identifying relationships**: the child entity does not need the parent entities identifier.



Dependent Entity

- When the child entity **does require attributes from the parent** entity to uniquely identify an instance.
- The child entity is called the dependent entity and its identifier consists of **at least one attribute from the parent entity**.
- Dependent entities are shown as a rectangle with **double border lines**.
- When relationships have a dependent child entity → **identifying relationship**.
- Intersectional entities are often dependent, unless they have a **logical identifier**. (e.g. transcript #)

